



Chest Tube Kit – 16Fr

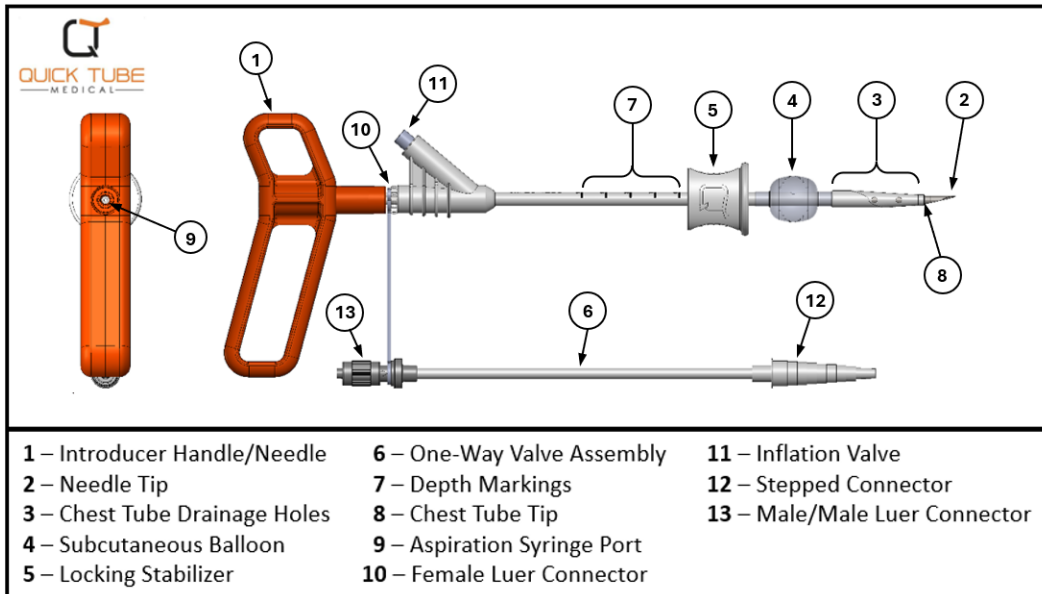


Figure 1: QT Chest Tube Kit

Note: Two (2) 10ml syringes are included, but not shown.

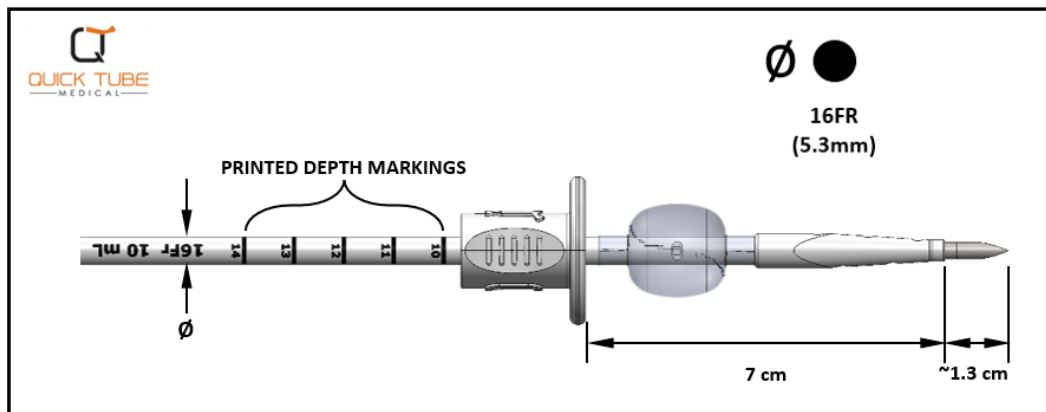


Figure 2: QT Chest Tube Dimensions (Not to Scale)

Note: The numbers on the chest tube shaft indicate the distance from the distal tip of the chest tube, measured in centimeters. The tip of the needle extends approximately 1.3 cm beyond the distal tip of the chest tube.

CONTENTS OF PACKAGE:

- (1) QT Chest Tube and delivery system
- (2) 10mL syringe

CAUTION

Federal (USA) Law restricts this device to sale by or on the order of a physician.

Do not use on pregnant or lactating patients.

DEVICE DESCRIPTION

The QT Chest Tube Kit (Figure 1) is a sterile, single use chest tube and delivery system. The Introducer Handle (1) contains a sharp, beveled needle tip (2) that allows percutaneous introduction of the chest tube to the pleural cavity without the need of a scalpel incision. After placement of the chest tube, the Introducer Handle/Needle is withdrawn and a subcutaneous balloon (4) is inflated. A Locking stabilizer (5) slides forward to secure the chest tube in place without the need for sutures. A one-way valve assembly (6) is then placed in line with, and is secured to, the chest tube. A commercially available suction device may be attached, if desired.

The QT Chest Tube is available in 16FR. Physicians should ensure the size of the device is appropriate for the specific procedure and patient. Printed graduation marks (7), measured in centimeters, indicate distance as measured from the distal tip of the chest tube (8) (see Figure 2.)

The patient contacting portions of the QT Chest Tube are constructed of radiopaque Pebax and Pellethane. The QT Chest Tube Kit does not contain Diethylhexyl phthalate (DEHP) or latex.

INTENDED USE / INDICATIONS FOR USE

The QT Chest Tube Kit is intended for percutaneous introduction and placement of the QT Chest Tube for the evacuation of air and/or fluids from the chest cavity such as may occur with pneumothorax, hemothorax, chylothorax, pleural effusion, or empyema.

CONTRAINDICATIONS

Relative contraindications to chest tube use may include pulmonary adhesions, lack of a pleural space, and hematologic abnormalities such as bleeding diatheses or coagulopathy.

WARNING

- Do not use after expiration date.
- Upon removal from package, inspect the product to ensure no damage has occurred. Do not use if the package is open or damaged.
- For single patient use only. Do not reuse, reprocess or re-sterilize. Reuse, reprocessing or re-sterilization may compromise the structural integrity of the device and/or lead to device failure which, in turn, may result in patient injury, illness, or death.

PRECAUTIONS

- This product is intended for use by physicians trained and experienced in the treatment of conditions requiring chest tube placement. Standard universal precautions for placement of chest tubes should be employed.
- Care should be taken during insertion, fixation and removal of the chest tube. Lung puncture may result in an air embolus, which could lead to ischemia or infarction of major organs, including the brain or cardiac system.
- The drainage holes on the distal portion of the chest tube must remain inside the thoracic cavity at all times during patient use.

HOW SUPPLIED

Supplied sterilized by ethylene oxide gas in peel-open packages. Intended for one-time use. Sterile if package is unopened or undamaged.

HANDLING AND STORAGE

Store in a dark, dry, cool place. Avoid extended exposure to light.

DEVICE PREPARATION AND USE

1. Prep, drape and sterilize the access site. The standard insertion site is the fourth intercostal space at the mid-axillary line.
2. Introduce a local anesthetic through the skin and subcutaneous tissue down to the pleura. **Anesthesia may be omitted in an emergency decompression.**
3. **Device prep.** Prior to use, execute the following:
 - a. Remove the plastic needle tip protector (not shown) to expose the balloon and sharp, beveled needle tip ②.
 - b. Complete a standard balloon functionality check by filling one (10 mL) syringe with 10 mL of air. Firmly attach this air-filled syringe to the Inflation Valve ⑪ and inflate the Subcutaneous Balloon ④ with 10 mL of air and deflate by pulling back on the plunger while maintaining pressure on the Inflation Valve ⑪.
4. Attach the second (10 mL, not shown) syringe to the Aspiration Syringe Port ⑨ located in the Introducer Handle/Needle ①.
5. Use the Introducer Handle to guide the needle tip through the skin, over the rib and into the pleural space.
6. The Depth Markings on the chest tube (per Figure 2) can be utilized to assist in controlling and confirming insertion depth.
7. Once inserted into the pleural space, begin pulling back on the plunger of the syringe attached to the Introducer Handle to aspirate air or fluid from the pleural space. If the Locking Stabilizer ⑤ meets the skin prior to successful aspiration, pinch the contoured sides of the Locking Stabilizer ⑤ to release it and then retract as needed to allow for more length of the device to be inserted.
8. Once aspiration has occurred, and the tip of the chest tube is confirmed to be within the pleural space, remove syringe from Introducer handle.
9. While gripping the hub of the chest tube with one hand, grip the Introducer Handle ① and twist counterclockwise to unscrew it from the hub of the catheter.
10. Withdraw the Introducer Handle/Needle ① 2-3 centimeters.
11. Pinch the Locking Stabilizer and slide it proximally to the hub of the Catheter, then feed the catheter further over the Introducer Needle into the pleural space. This ensures the balloon is situated fully beyond the chest wall prior to inflation.
12. Once the Catheter is advanced sufficiently into the pleural space, continue to withdraw the Introducer until the needle fully exits the chest tube.
13. Fill the first (10 mL) syringe with 10 mL of air. Firmly attach this air-filled syringe to the Inflation Valve ⑪ and inflate the Subcutaneous Balloon ④ with 10 mL of air. While maintaining pressure on the plunger of the syringe, quickly remove the syringe from the inflation valve. Gently pull back on the chest tube to confirm that the balloon is holding the chest tube in place within the pleural cavity.
14. Place petrolatum gauze around the insertion site, between the Locking Stabilizer and the patient's skin.
15. Pinch and slide the Locking Stabilizer forward to position it firmly against the petrolatum gauze. Release the Locking Stabilizer to secure the chest tube in place.
16. Position the One-Way Valve Assembly ⑥ in-line with the chest tube and secure the parts together by connecting the Male/Male Luer Connector ⑬ to the Luer Fitting ⑩.
17. The Stepped Connector ⑫ may then be connected to a commercially available water seal suction apparatus or similar suction device, if desired.
18. While chest tube is in place, maintain a sterile occlusive dressing per standard fashion.
19. Replace the tip protector to cover the exposed sharp beveled needle tip ② or dispose directly into a suitable sharps container.

VERIFICATION OF TUBE PLACEMENT

Confirm device placement using a standard chest radiograph or with fluoroscopy. The chest tube is radiopaque to allow visualization of the distal tip and main shaft.

DEVICE MAINTENANCE

Chest Tube may remain in place for up to 30 days.

Confirm the device securement every 72 hours, at a minimum, by lightly tugging on the device to ensure the balloon remains fully inflated. Add a couple of ml of air to the balloon, as needed.

Avoid excessive tension on chest tube or connections.

If needed, the one-way valve assembly ⑥ may be detached to allow the attachment of a syringe to the Luer fitting ⑩ for flushing of the chest tube.

DEVICE REMOVAL

To remove the chest tube:

- 1. Firmly attach an empty 10mL (or larger) syringe to the inflation valve ⑩.
- 2. Pull back the syringe plunger to deflate the Subcutaneous Balloon ④.
- 3. Have the patient inhale and hold a deep breath as the Chest Tube is quickly withdrawn.
- 4. Apply a sterile occlusive dressing.

DEVICE DISPOSAL

Dispose of Device in accordance with local and Federal regulations.

MR SAFETY INFORMATION

 MRI Safety Information A person with Quick Tube's Chest Tube Device may be safely scanned at 1.5T or 3.0T under the following conditions. Failure to follow these conditions may result in injury.	
Parameter	Condition
Device Name	Quick Tube's Chest Tube Device
Static Magnetic Field Strength (B0)	1.5T and 3.0T
MR Scanner Type	Cylindrical
B0 Field Orientation	Horizontal
Maximum Spatial Field Gradient	20 T/m (2,000 Gauss/cm)
RF Excitation	Circularly Polarized (CP)
RF Transmit Coil Type	Integrated Whole Body Transmit Coil
Operating Mode	Normal Operating Mode
Scan Duration	2 W/kg whole-body average SAR for 60 minutes of continuous RF (a sequence or back to back series/scan without breaks)
Image Artifact	The presence of Quick Tube's Chest Tube Device may produce an image artifact external to the patient body. Some manipulation of scan parameters may be needed to compensate for the artifact.

SYMBOLS GLOSSARY

 DO NOT USE IF PACKAGE IS DAMAGED	 CAUTION	 USE BY
 DATE OF MANUFACTURE	 CATALOGUE NUMBER	 KEEP DRY
 BATCH CODE	 STERILIZED USING ETHYLENE OXIDE	 CONSULT INSTRUCTIONS FOR USE
 KEEP AWAY FROM SUNLIGHT	 DO NOT RESTERILIZE	 DO NOT REUSE
 Rx ONLY (USA)	 MR CONDITIONAL	

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